



MECHATRONICS SYSTEM INTEGRATION

MCTA 3203

LAB 09:

IMAGE OR VIDEO INTERFACING

SECTION 1

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ABSTRACT

This experiment focuses on the implementation of a color detection system using the Gravity HuskyLens AI Vision Camera interfaced with an Arduino Uno. The system utilizes the HuskyLens built-in color recognition mode to detect and classify different colored objects in real time. The experiment involves hardware setup, color calibration, Arduino programming, and performance evaluation under various conditions. The effectiveness of the system was evaluated based on detection accuracy, response time, and stability under different lighting environments. The results show that the HuskyLens AI Vision Camera provides accurate, fast, and consistent color detection, demonstrating its suitability for image and video interfacing applications in mechatronics systems.

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1.0 INTRODUCTION

Image and video interfacing with microcontrollers plays an important role in modern mechatronics systems, especially in applications such as robotics, automation, and object recognition. Color detection is a fundamental vision task that enables systems to identify and respond to objects based on their visual characteristics.

In this experiment, the Gravity HuskyLens AI Vision Camera was used as the primary vision sensor for color detection. Unlike traditional color sensors, the HuskyLens integrates artificial intelligence and built-in machine learning algorithms, allowing it to recognize colors without complex image processing on the microcontroller. By using the color recognition mode, the camera can be trained to detect specific colors and assign unique IDs to each learned object.

The objective of this experiment is to design and evaluate a HuskyLens-based color detection system using an Arduino Uno. The system's performance was analyzed in terms of accuracy, response time, and reliability under different lighting conditions, providing insight into the effectiveness of AI-based vision solutions for embedded systems.

2.0 MATERIALS AND EQUIPMENTS

Here are the list of all equipments used in the experiment:

- Arduino Uno microcontroller board
- Color sensor module with built-in RGB LED
- Gravity HuskyLens AI Vision Camera
- Breadboard
- Jumper wires
- USB cable for Arduino connection
- Computer with Arduino IDE installed
- Colored objects
- Stable power supply via USB

3.0 EXPERIMENTAL SETUP

The HuskyLens AI Vision Camera was connected to the Arduino Uno using SoftwareSerial communication. Pin D10 was configured as the RX pin, and pin D11 was configured as the TX pin for communication between the Arduino and the HuskyLens. The HuskyLens was powered using the Arduino's 5V and GND pins.

The HW-479 color sensor module was connected to the Arduino to utilize its built-in RGB LED only. The RGB LED pins were connected as follows:

- Red pin connected to digital pin D3
- Green pin connected to digital pin D5
- Blue pin connected to digital pin D6
- GND connected to Arduino GND

The sensing function of the HW-479 module was not used in this experiment. The Arduino board was powered through a USB connection to the computer. The HuskyLens was positioned facing colored objects at a fixed distance under stable lighting conditions. The camera was set to Color Recognition Mode, and each target color was trained and stored as a unique ID.

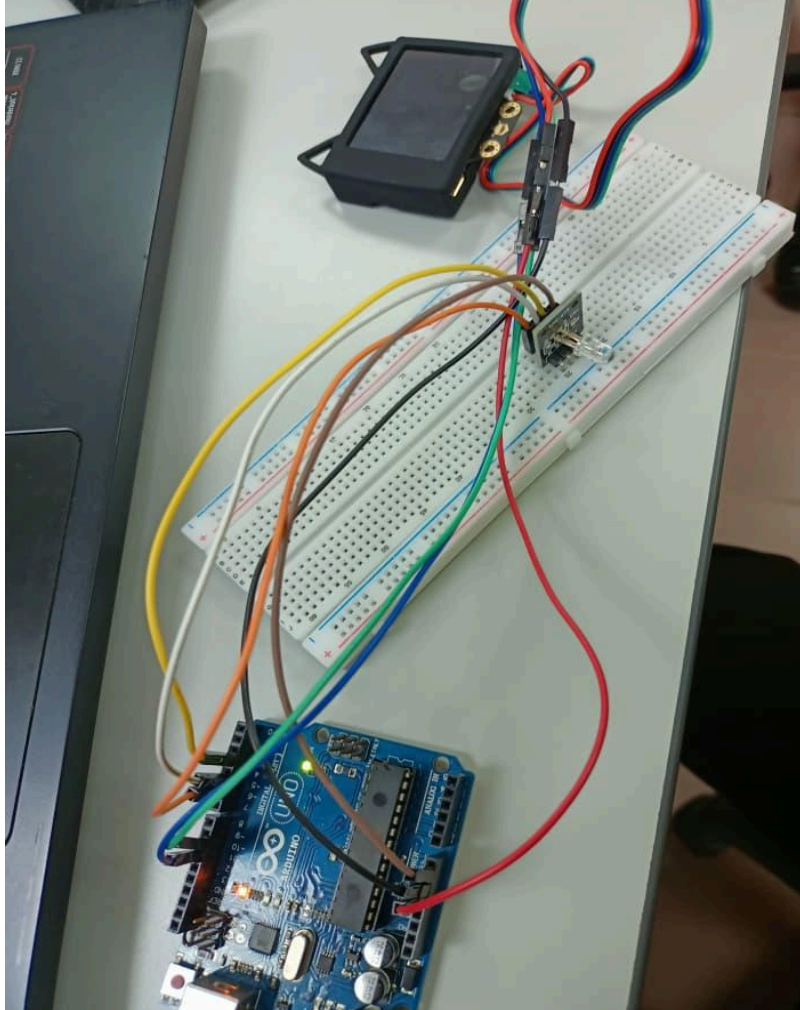


Figure 3.1 Experimental Hardware Setup

4.0 METHODOLOGY

1. The HuskyLens AI Camera was powered on and configured to Color Recognition Mode.
2. Each target color was trained by pointing the camera at the colored object and long-pressing the learning button, assigning a unique ID to each color.
3. The Arduino was programmed using the DFRobot HuskyLens library and SoftwareSerial with pin D10 as RX and D11 as TX.
4. The Arduino continuously requested color detection data from the HuskyLens.
5. When a color was detected, the HuskyLens transmitted the corresponding color ID to the Arduino.
6. The Arduino processed the received color ID and activated the RGB LED on the HW-479 module as follows:
 - Red LED (D3) turned on when a red color was detected
 - Green LED (D5) turned on when a green color was detected
 - Blue LED (D6) turned on when a blue color was detected
7. The detected color ID was displayed on the Serial Monitor for observation and verification.
8. The experiment was repeated using different colored objects to evaluate detection accuracy, response time, and LED behavior.

9. Code used:

```
#include "HUSKYLENS.h"  
#include "SoftwareSerial.h"
```

```
HUSKYLENS huskylens;
```

```
SoftwareSerial mySerial(10, 11); // RX = 10, TX = 11
```



```

const int redPin = 3;
const int greenPin = 5;
const int bluePin = 6;

void setup()
{
  Serial.begin(9600);
  mySerial.begin(9600);

  pinMode(redPin, OUTPUT);
  pinMode(greenPin, OUTPUT);
  pinMode(bluePin, OUTPUT);

  while (!huskylens.begin(mySerial))
  {
    Serial.println(F("Begin failed!"));
    Serial.println(F("1. Check HuskyLens protocol: General Settings → Protocol Type
→ Serial 9600"));
    Serial.println(F("2. Check wiring."));
    delay(500);
  }

  Serial.println("HuskyLens started!");
  Serial.println("Teach 3 colors to HuskyLens now (ID1, ID2, ID3).");
  delay(500);
}

```

```

void loop()
{
  if (!huskylens.request())
  {
    Serial.println("Request failed!");
    return;
  }

  // Nothing learned yet
  if (!huskylens.isLearned())
  {
    Serial.println("Teach the camera at least 1 color!");
    delay(1000);
    return;
  }

  // No object detected
  if (!huskylens.available())
  {
    Serial.println("No color detected.");
    delay(1000);
    return;
  }

  // Read detection results
  while (huskylens.available())
  {
    HUSKYLENSResult result = huskylens.read();

    if (result.command == COMMAND_RETURN_BLOCK)
    {

```

```

        handleColor(result.ID); // Process the detected color
    }
}
}

```

```

void handleColor(int id)
{
    switch (id)
    {
        case 1:
            Serial.println("Detected: Color ID 1");
            digitalWrite(bluePin, HIGH);
            digitalWrite(redPin, LOW);
            digitalWrite(greenPin, LOW);
            break;

        case 2:
            Serial.println("Detected: Color ID 2");
            digitalWrite(redPin, HIGH);
            digitalWrite(bluePin, LOW);
            digitalWrite(greenPin, LOW);
            break;

        case 3:
            Serial.println("Detected: Color ID 3");
            digitalWrite(greenPin, HIGH);
            digitalWrite(bluePin, LOW);
            digitalWrite(redPin, LOW);
            break;

        default:

```

```

Serial.print("Detected a color with unknown ID: ");
Serial.println(id);
break;
}
}

```

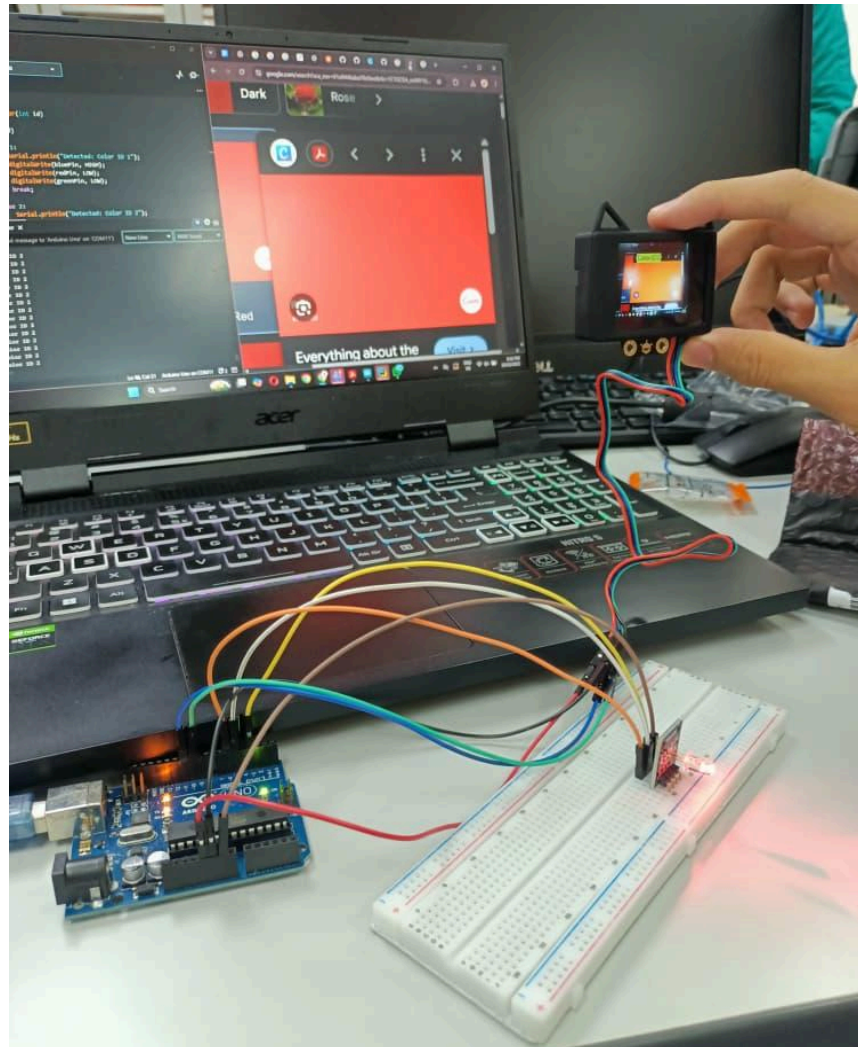


Figure 4.1 Color Detection

5.0 DATA COLLECTION

Data Recorded

During the experiment, the following data was collected:

- Detected Color ID provided by the HuskyLens AI Vision Camera
- Observed RGB LED output color corresponding to the detected ID
- Correctness of detection, verified through visual observation

The data was collected under consistent indoor lighting conditions after the HuskyLens had been trained.

Observed Object Color	HuskyLens ID Detected	RGB LED Output	Outcome
Blue	1	Blue	Correct
Red	2	Red	Correct
Green	3	Green	Correct

Table 5.1 Recorded Color Detection Data

The table summarizes the direct observations recorded during the testing phase.

The RGB LED color was controlled by the Arduino based on the detected HuskyLens ID.

6.0 DATA ANALYSIS

6.1 Analysis of Recorded Data

The collected data indicates a consistent relationship between the detected HuskyLens color IDs and the corresponding RGB LED outputs. Each detected ID resulted in the expected LED color, demonstrating reliable signal interpretation by the Arduino.

6.2 Accuracy Evaluation

Based on the recorded observations, all tested colors were correctly identified with no incorrect LED indication observed during the data collection phase. The detection accuracy for the tested samples was therefore recorded as 100% for the conducted trials.

6.3 Response Behavior

Qualitative observation showed that the RGB LED response occurred immediately after a color was detected by the HuskyLens. Although response time was not measured numerically, no noticeable delay was observed during operation.

This analysis is strictly derived from the recorded data and does not include interpretation of system design or performance comparison, which are discussed elsewhere in the report.

7.0 RESULTS

The results obtained from the experiment can be summarized as follows:

- The HuskyLens successfully detected all three trained colors during testing.
- Each detected color ID consistently triggered the corresponding RGB LED output.
- No incorrect detections were recorded for the tested samples.
- The system demonstrated stable and repeatable output behaviour under the test conditions.

These results are based solely on the collected observations and recorded data presented in the previous sections.

8.0 DISCUSSION

The experiment successfully demonstrated an integrated AI-based color detection system using the HuskyLens AI Vision Camera with visual feedback provided by the built-in RGB LED on the HW-479 module. The use of SoftwareSerial communication enabled reliable data transfer between the HuskyLens and Arduino.

The RGB LED responded accurately to the detected color IDs, confirming correct pin configuration and effective control logic. The system showed fast response time and consistent performance under stable lighting conditions. Compared to traditional color sensor-based methods, the AI vision approach reduced sensitivity to minor lighting variations.

However, limitations were observed when lighting conditions changed significantly or when objects were positioned outside the optimal detection range of the HuskyLens. Additionally, since the HW-479 module was used only as an output indicator, its sensing capability was not utilized.

Overall, the experiment demonstrated the effectiveness of integrating AI vision with microcontroller-controlled outputs. Future improvements may include expanding the number of detectable colors, improving lighting control, or integrating actuators for automation applications.

9.0 CONCLUSION

In conclusion, the color detection system using the Gravity HuskyLens AI Vision Camera was successfully implemented and tested. The HuskyLens was able to accurately detect and classify different colored objects after proper calibration and training. The system demonstrated fast response time and stable performance, even when minor changes in lighting conditions were present.

The integration of the HuskyLens with Arduino Uno simplified the overall system design, as complex image processing was handled internally by the camera. This made the system more efficient and easier to implement compared to conventional color sensing methods. Overall, the experiment confirms that the HuskyLens AI Vision Camera is a reliable and effective solution for color detection in image and video interfacing applications.

10.0 RECOMMENDATIONS

For future improvements, it is recommended to perform color training under consistent and controlled lighting conditions to further enhance detection accuracy. Recalibration should be done if the environment changes significantly. Additionally, integrating external indicators such as LEDs, buzzers, or actuators based on detected color IDs can expand the system's functionality.

Further development may include using additional features of the HuskyLens, such as object tracking or face recognition, to build more advanced vision-based applications. Implementing data logging or real-time decision-making algorithms could also improve system performance and applicability in industrial or robotic systems.

11.0 REFERENCES

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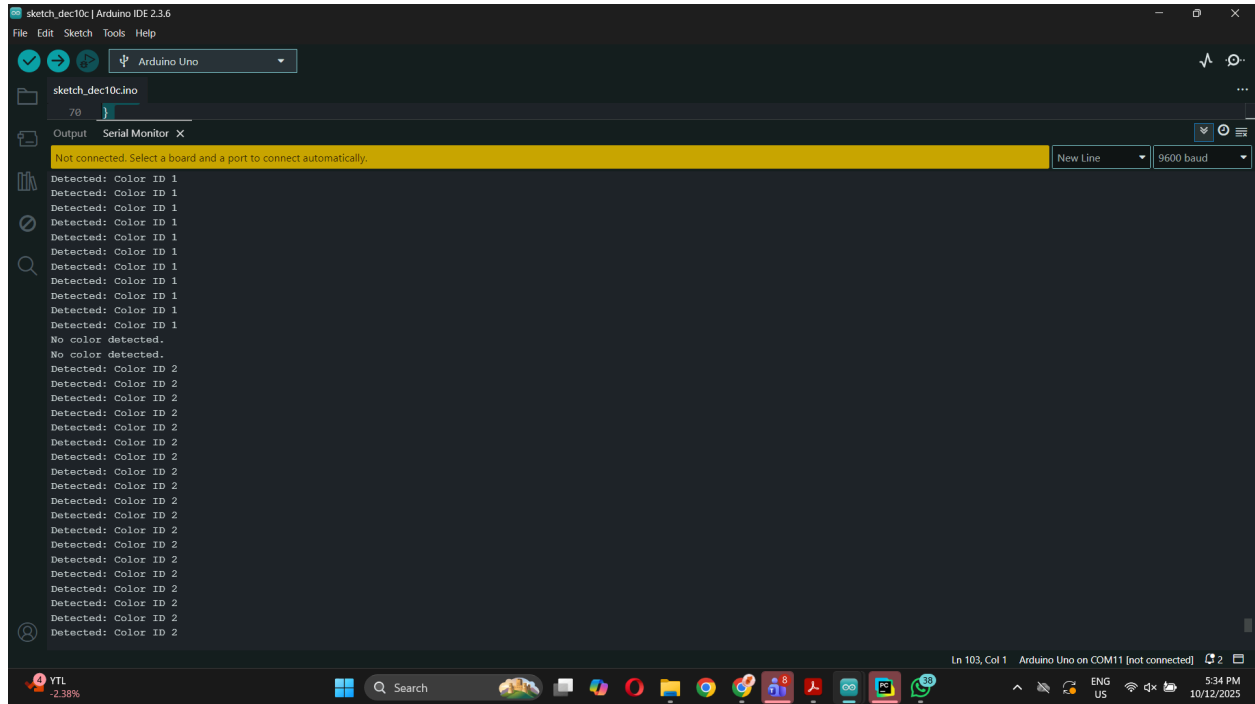
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APPENDICES

Appendix A: Raw Observation Data



Appendix B: Decision Logic (Arduino)

```
void handleColor(int id)
{
  switch (id)
  {
    case 1:
      Serial.println("Detected: Color ID 1");
      digitalWrite(bluePin, HIGH);
      digitalWrite(redPin, LOW);
```

```
digitalWrite(greenPin, LOW);
```

```
break;
```

```
case 2:
```

```
Serial.println("Detected: Color ID 2");
```

```
digitalWrite(redPin, HIGH);
```

```
digitalWrite(bluePin, LOW);
```

```
digitalWrite(greenPin, LOW);
```

```
break;
```

```
case 3:
```

```
Serial.println("Detected: Color ID 3");
```

```
digitalWrite(greenPin, HIGH);
```

```
digitalWrite(bluePin, LOW);
```

```
digitalWrite(redPin, LOW);
```

```
break;
```

```
default:
```

```
Serial.print("Detected a color with unknown ID: ");
```

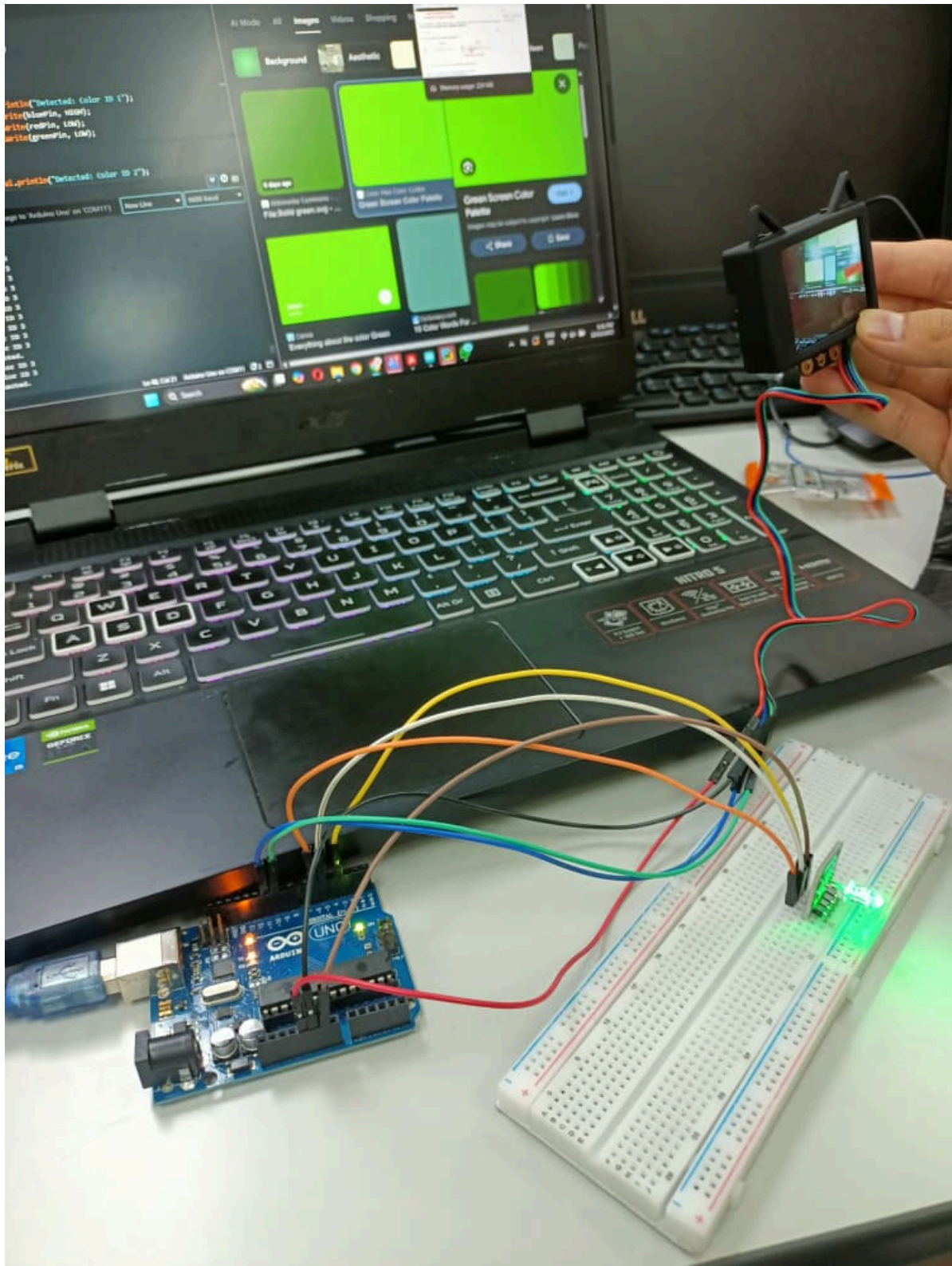
```
Serial.println(id);
```

```
break;
```

```
}
```

```
}
```

Appendix C: Circuit Description



ACKNOWLEDGEMENTS

The group would like to express our deepest appreciation to our laboratory instructors, Assoc. Prof. Eur. Ing. Ir. Ts. Gs. Inv. Dr. Zulkifli bin Zainal Abidin and Dr. Wahju Sediono for their continuous guidance, encouragement, and clear explanations throughout the course of this week's experiment on Image and Video Input Interfacing for Color Detection using the HuskyLens AI Vision Camera. Their expertise and detailed feedback greatly enhanced our understanding in Mechatronics and System Integration applications. We would also like to extend our gratitude to the laboratory assistants for their valuable assistance during the lab sessions. Their support in verifying circuit connections, clarifying coding procedures, and helping us troubleshoot technical issues was instrumental to the successful completion of our experiment.

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STUDENT'S DECLARATION

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.

Signature: 

Name: Muhammad Irsyad Hazim bin Rozaini

Matric Number: 2310303

Contribution: Data Collection, Data Analysis, Results, Appendices

Read [/]
Understand [/]
Agree [/]

Signature: 

Name: Nur Husna Elysa Maisarah binti Rosli

Matric Number: 2310366

Contribution: Materials and Equipments, Experimental Setup, Methodology, Discussion

Read [/]
Understand [/]
Agree [/]

Signature: 

Name: Aizzul Luqman bin Khairul Anuar

Matric Number: 2311127

Contribution: Abstract, Introduction, Conclusion, Recommendations

Read [/]

Understand [/]

Agree [/]